

UTD Weekly Problem, 10-1-16

- Point your browser at <https://pc2team.utdallas.edu/pc2team>
- Use a team name and login/password in [team20,team100]. If someone is using it, try another one.
- Use only Java, C or C++.
- Your program source code must be contained in one source file.
- Do not use the “package” construct in your Java code.
- Your programs must read from the data file given at the top of the problem description (none in this problem) and output to System.out (cout for C/C++ programmers.)
- Make sure that the last line of your output ends in a newline (i.e. is printed with System.out.println(...))
- **Do not copy solutions from the web.**
- As soon as you have solved a problem, send me an email containing your name and your team name.
- The deadline is Friday 10-7-16 at midnight.
- The first person to submit a correct solution wins \$10 etc.

[A] Quarters

Data File = There is no data file for this problem
Time limit = 10 seconds

Four Quarters is a game of chance played with four quarters. Two people, called A and B, each flip two quarters each round. They each gain or lose points each round based on the following tables:

		Player B		
		HH	HT	TT
Player A	HH	1	1	2
	HT	0	0	1
	TT	-1	0	0

Player A's payoff

		Player B		
		HH	HT	TT
Player A	HH	0	-1	-1
	HT	1	0	0
	TT	2	0	-1

Player B's payoff

There is no difference between Heads/Tails and Tails/Heads. As you can see, the odds are stacked in Player A's favor. At the beginning of the game, each player has 0 points, and points accumulate as the game progresses. At the end of the game, whichever player has the most points wins. You must write a program that determines the probability that Player A will win, Player B will win, or they will tie, after a certain number of rounds. Assume that the coins are fair, i.e. that heads and tails are equally likely.

Input:

There is no input file for this problem.

Output:

Output a table that lists the probability that Player A will win, B will win, or they will tie, after each round for 1 to 20 rounds. The output for rounds 1 through 3 is given below.

Probabilities must be expressed as a percent, with 4 places after the decimal.

Round	A wins	B wins	Tie
1	43.7500%	18.7500%	37.5000%
2	56.6406%	22.2656%	21.0938%
3	62.3535%	22.7051%	14.9414%